

Introduction:

1-Project Definition:

This project focuses on the design and implementation of an Automated Industrial Elevator Control System using a Programmable Logic Controller (PLC). In modern industrial facilities and warehouses, elevators are critical for the vertical transport of heavy materials and equipment. This system replaces traditional hard-wired logic with a flexible, software-driven approach to manage floor requests, motor control, and safety protocols.

2-Technical Advantage:

Using a PLC for elevator control provides a significant advantage in Troubleshooting and Maintenance. Unlike manual systems, the PLC allows for real-time monitoring of every sensor's status. Furthermore, safety features like Auto-Return to Home or Overload Detection can be easily programmed and updated without physical hardware changes.

3-System Components:

This section outlines the hardware and simulated industrial components that comprise the multi-level lifting and sorting automation cell. The system is divided into three primary categories: Actuators (Outputs), Sensors (Inputs), and Control & Display Units.

1. Actuators & Driving Mechanisms (Outputs):

These components execute the mechanical movements commanded by the Siemens S7-1500 PLC:

Entry Conveyor (%Q0.0): A heavy-duty motor-driven friction conveyor that handles the initial feeding and queuing of incoming boxes into the cell.

Left Lift Vertical Drive (%Q0.7, %Q1.0, %Q1.1): A bi-directional, dual-speed induction motor assembly capable of rapid hoisting (Left up), precise micro-positioning leveling (Left slow), and rapid lowering (Left down).

Right Lift Vertical Drive (%Q1.2, %Q1.3, %Q1.4): A secondary vertical hoisting system mirror-configured to the left lift, executing high-speed lifting (Right up), crawling speed adjustment (Right slow), and rapid lowering (Right down).

Onboard Deck Conveyors (%Q0.2, %Q0.5): Bi-directional roller motorized tracks mounted directly inside the elevator cabins (Load left and Load right) responsible for drawing boxes inward from the tracks and discharging them onto the sorting levels.

Horizontal Distribution Conveyors (%Q0.3, %Q1.5, %Q1.6): Lateral transit tracks assigned across the three floors (Conveyor 0 for Ground, Conveyor 1 for Level 1, and Conveyor 2 for Level 2) that bridge the two lifts together.

Exit Conveyor (%Q0.6): The final motorized transport belt that clears processed and sorted boxes out of the system cell.

2. Sensing & Feedback Instrumentation (Inputs):

These components provide the real-time physical telemetry required for state-machine execution:

Arrival & Exit Photo-eyes (%I10.0, %I10.7): High-precision photoelectric retro-reflective sensors used to log throughput data (At entry and At exit) and feed the counting logic blocks.

Cabin Presence Sensors (%I10.1, %I10.5): Inductive/photoelectric proximity sensors located inside the moving lift carriages (At entry left and At entry right) to dynamically confirm whether a lift is holding cargo or empty.

Floor Boundary Tracking Limits (%I11.4 to %I12.7): A network of physical limit switches and proximity trackers positioned at exact vertical intervals along the hoistways (At left low 0/1/2, At left high 0/1/2, At right low 0/1/2, and At right high 0/1/2) to establish absolute position tracking.

3. Control, Human-Machine Interface (HMI) & Displays:

The infrastructure managing user oversight and systemic safety parameters:

Operator Control Panel (%I3.0, %I3.2, %I3.3): The physical or simulated pilot devices including the system latching toggle (Start), standard operational shutdown (Stop), and emergency structural isolated trip (Emergency stop).

Visual Status Indicators (%Q3.2, %Q3.4, %Q1.7, %Q2.0): Pilot lamps (Start light, Stop light) along with distinct audible/visual sirens (Left warning, Right warn) deployed to provide automated warnings during active vertical mechanical travel.

Multi-Segment LED Digital Screens (%QD30, %QD34): Industrial numeric double-word display units (digital display 0 and digital display 1) that display real-time calculated metrics regarding the values of total entered and successfully cleared boxes.

Industrial Elevator Controller

System Objective, Functional Architecture, and Technical Justification

1. System Objective

The primary objective of the control system is to manage, coordinate, and automate the vertical and horizontal transit operations within the material handling cell with absolute safety and optimal throughput efficiency. The system is engineered to seamlessly process transport requests initiated across different sorting floors and lift stations, routing cargo to its designated destinations in the shortest cycle time possible. Simultaneously, it enforces deterministic safety interlocks to prevent mechanical failure, cross-collisions, and workplace hazards.

2. System Function

The Programmable Logic Controller (PLC) acts as the central intelligence of the industrial lifting cell. Its functional execution loop is based on the continuous acquisition of field instrumentation telemetry to drive discrete and digital actuation sequences as detailed below:

- **Request Processing and Queuing:** Continuously scans and logs discrete inputs from floor proximity limit switches, photoelectric cells, and physical command buttons to register transport requests.
- **Directional and Motion Control:** Dynamically calculates the optimal vector of vertical transit (hoisting vs. lowering) and horizontal movement based on the real-time localized tracking of the carrier cabins relative to the target floor.
- **Gate and Deck Integration:** Controls the automated entry and discharge doors or internal motorized roller decks upon accurate level alignment, ensuring immediate runtime interruption if optical safety sensors or photocells detect an obstruction.
- **Precision Position Tracking:** Monitors the instantaneous physical position of the elevator cabins using high-precision inductive proximity sensors and hardwired limit switches to maintain absolute level boundaries and prevent over-travel faults.

3. Structural Justification: Why Use a PLC?

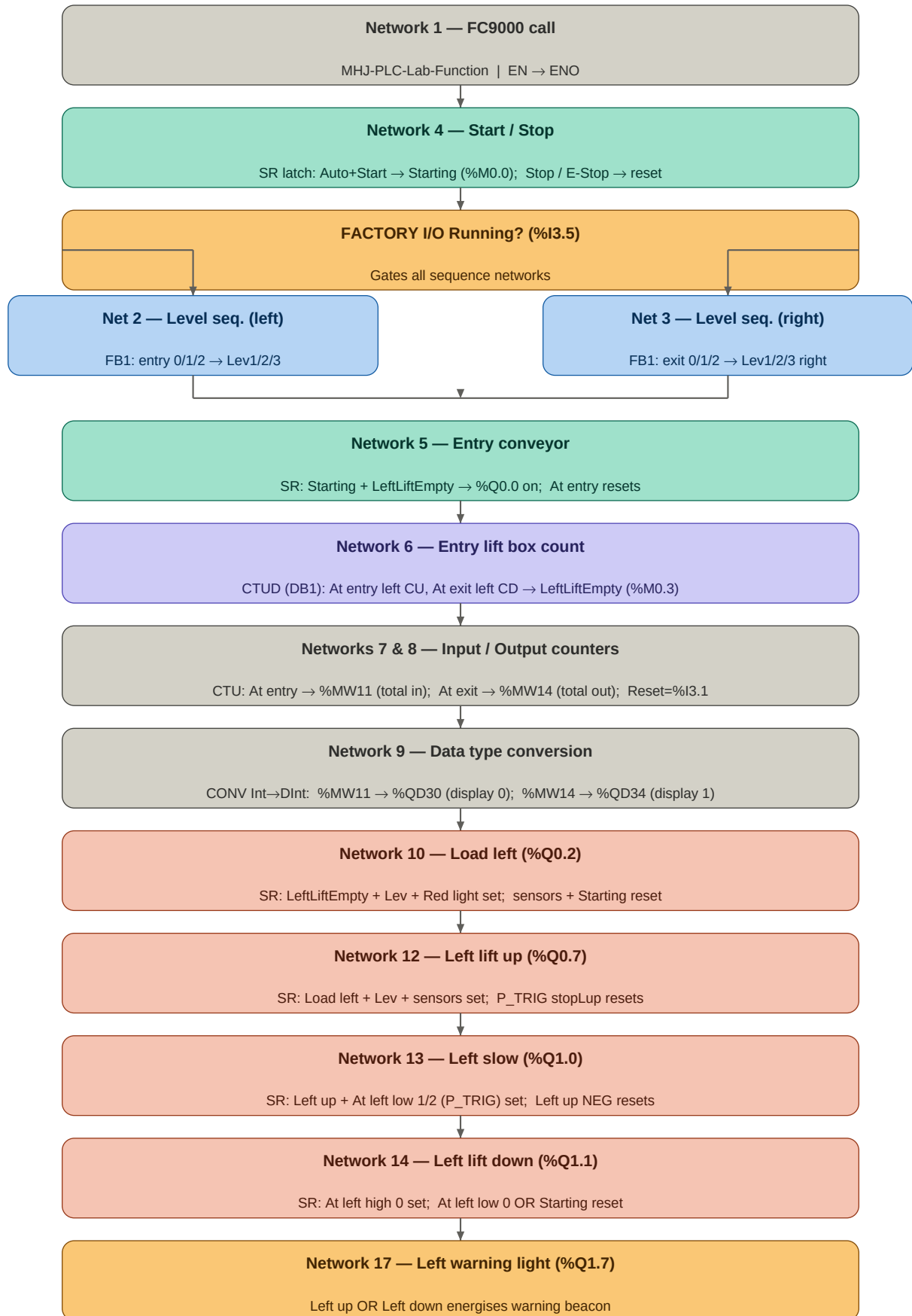
Historically, multi-level material handling mechanisms were driven by complex hardwired magnetic relay panels and extensive analog networks. This project replaces classical hardwired architectures with a solid-state Programmable Logic Controller (PLC) due to several distinct engineering advantages:

- **Industrial-Grade Reliability:** The PLC architecture is designed natively to operate continuously in harsh manufacturing environments, demonstrating high resilience against thermal fluctuations, electromagnetic interference (EMI), mechanical shocks, and vibration, which typically cause mechanical fatigue and contact degradation in standard relay systems.
- **Operational Flexibility and Scalability:** System modifications—such as adjusting conveyor dwell times, fine-tuning acceleration curves, or changing sorting sequences—are implemented strictly through soft-code modifications within the Ladder Logic program (TIA Portal). This eliminates the need for expensive physical rewiring, hardware re-allocation, or systemic downtime.

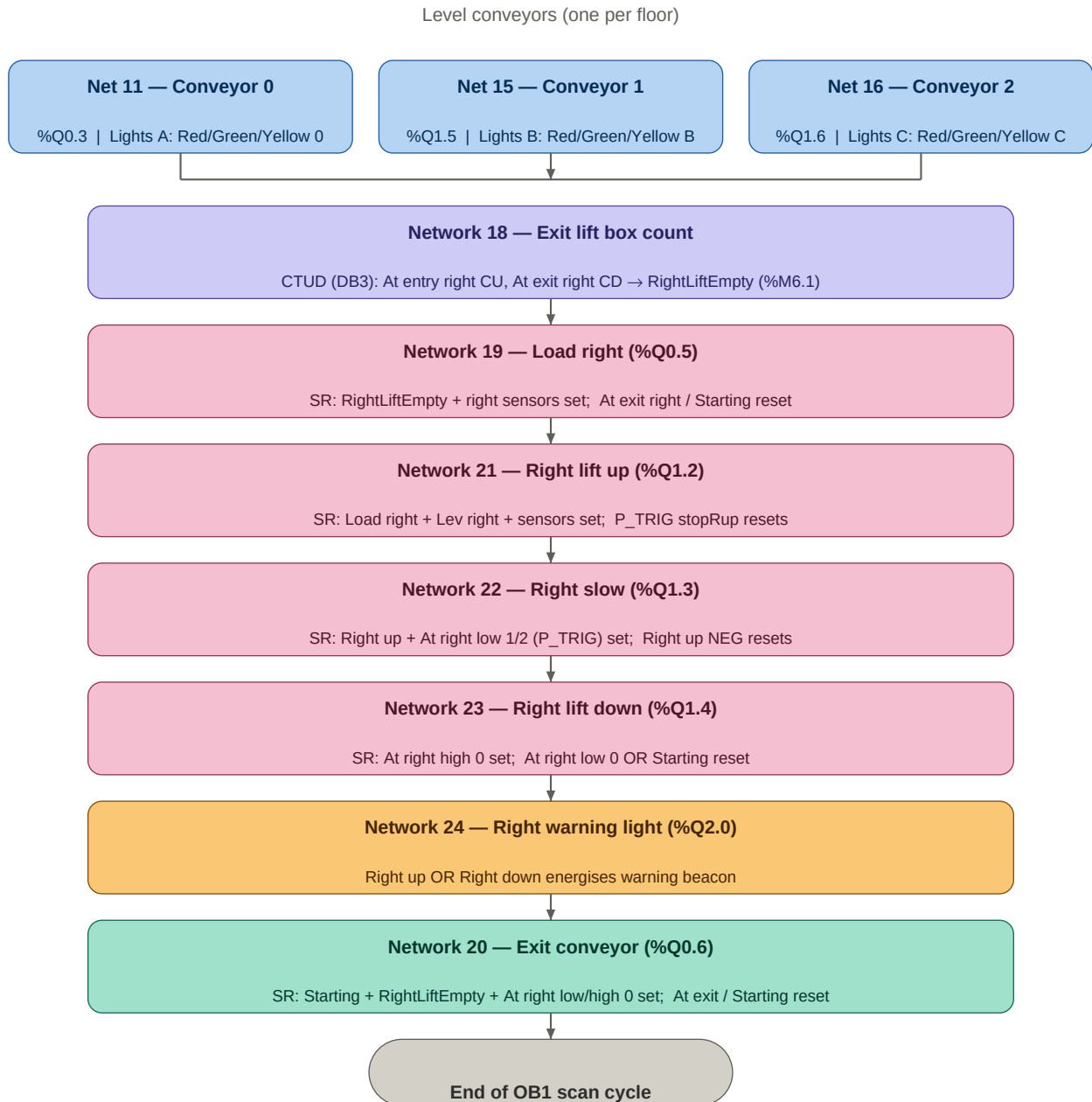
- **Footprint Minimization and Energy Efficiency:** Integrating all sequence variables into a single solid-state controller significantly reduces the physical dimensions of the control enclosure and lowers parasitic power consumption compared to large, heat-generating relay racks.
- **Advanced Fault Diagnostics and Troubleshooting:** The controller maintains constant monitoring over its local and distributed I/O image tables. Integrated LED status indicators and diagnostic data blocks allow engineering personnel to isolate field sensor faults, wire breaks, or actuator failures instantly, minimizing the Mean Time to Repair (MTTR).

FactoryIO Elevator — PLC Main OB1 Flowchart

Siemens S7-1500 | CPU 1511-1 PN | LAD (Ladder Diagram)



FactoryIO Elevator — Main OB1 (continued)



- | Legend | |
|---|---|
| ■ Start/stop control & conveyors | ■ Box detection counters (CTUD) |
| ■ Left lift motion (up / slow / down) | ■ Warning lights & FIIO gate |
| ■ Right lift motion (up / slow / down) | ■ Utility: FC call, counters, conversion |
| ■ Level sequencing & floor conveyors | |

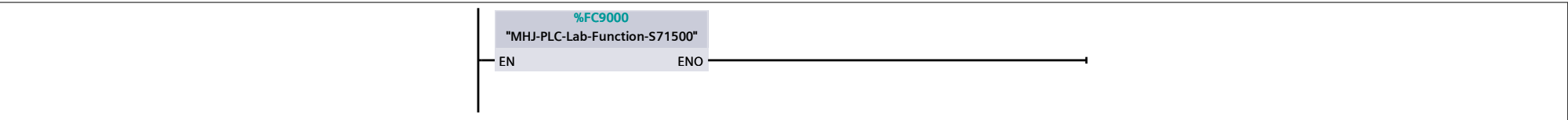
FactoryIO_Elevator mod / PLC_1 [CPU 1511-1 PN] / Program blocks

Main [OB1]

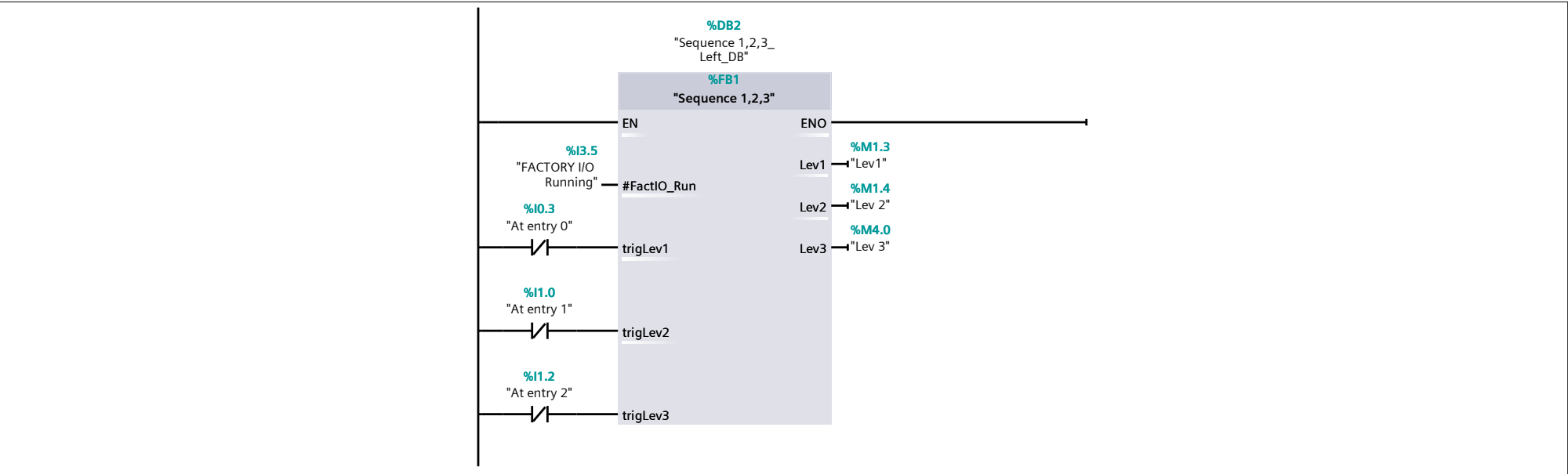
Main Properties							
General							
Name	Main	Number	1	Type	OB	Language	LAD
Numbering	Automatic						
Information							
Title	"Main Program Sweep (Cycle)"	Author		Comment		Family	
Version	0.1	User-defined ID					

Main			
Name	Data type	Default value	Comment
▼ Input			
Initial_Call	Bool		Initial call of this OB
Remanence	Bool		=True, if remanent data are available
Temp			
Constant			

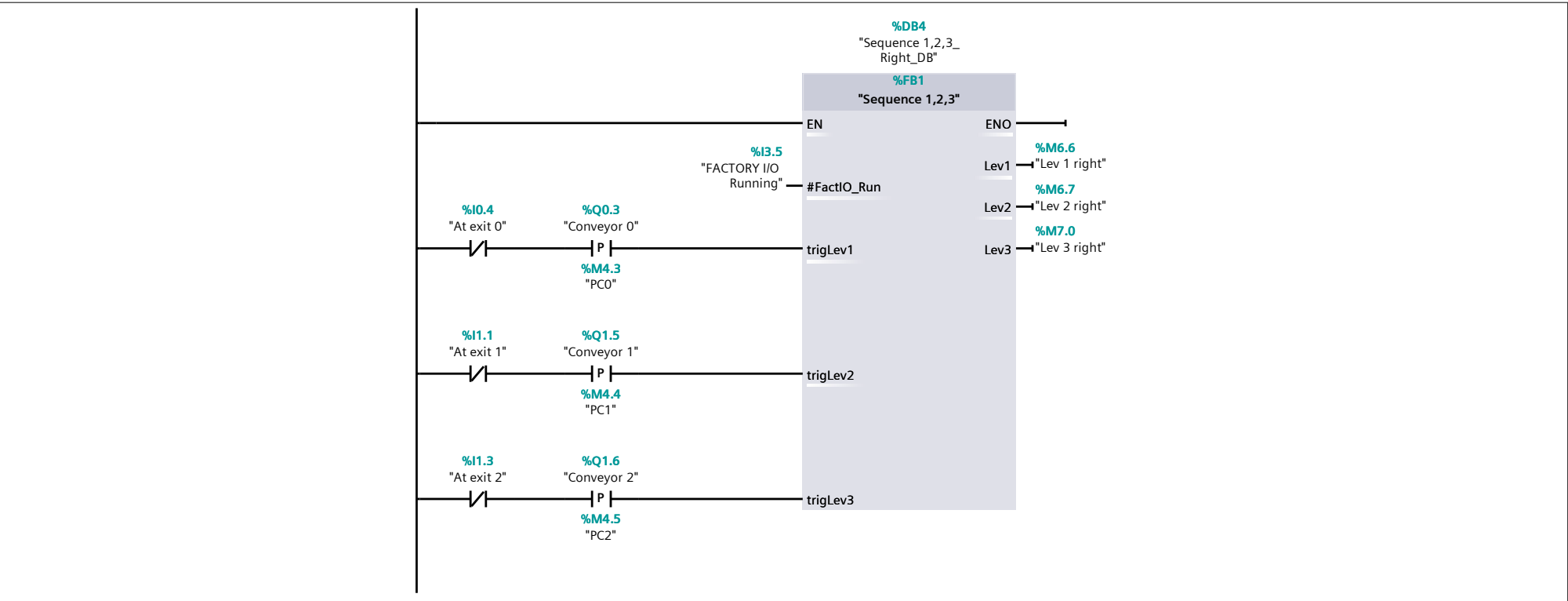
Network 1:



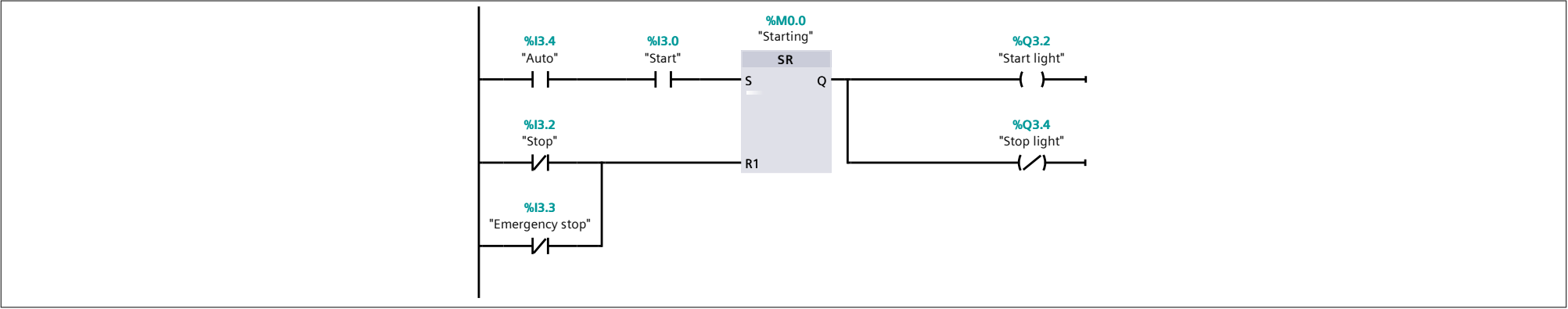
Network 2: Sequence of levels (left)



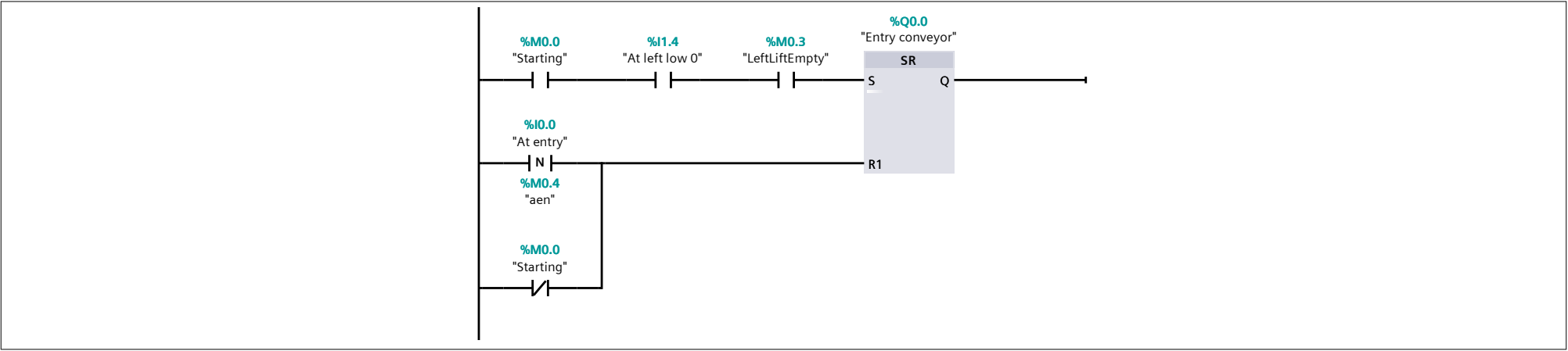
Network 3: Sequence of levels (right)



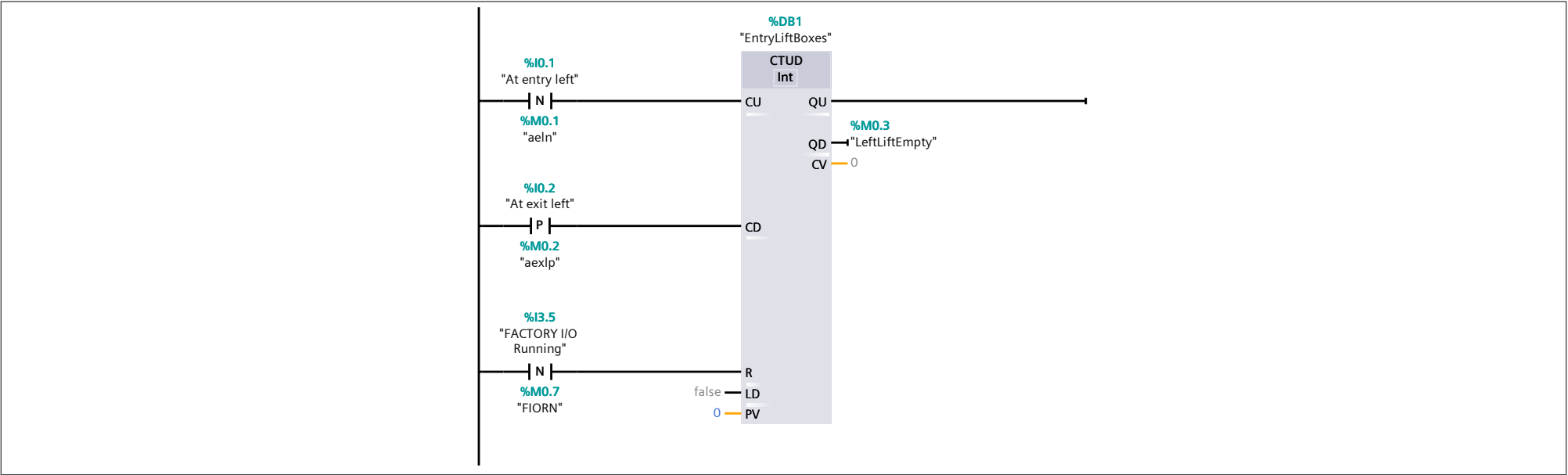
Network 4: Start/Stop



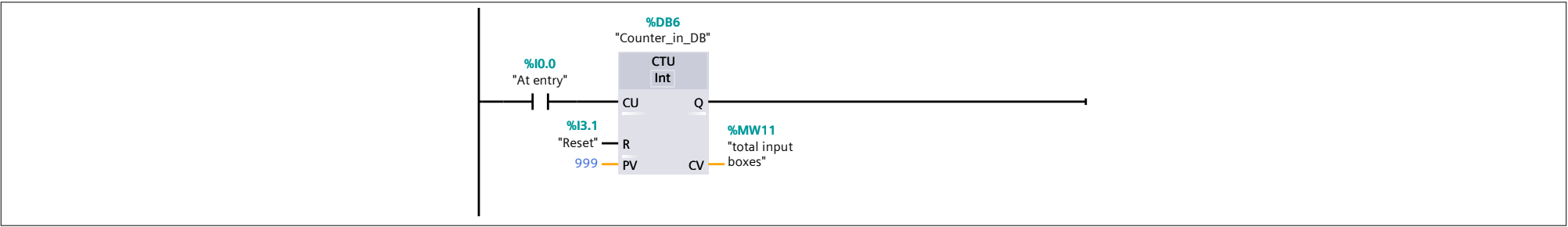
Network 5: Entry Conveyor



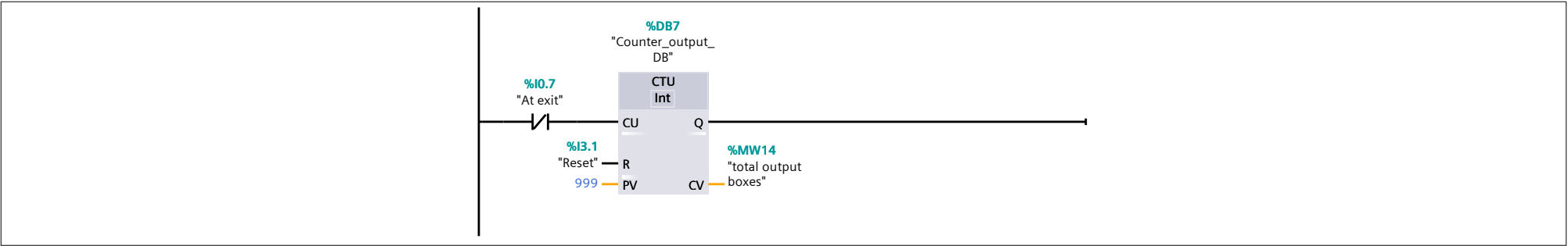
Network 6: Detecting presence of a box on entry lift



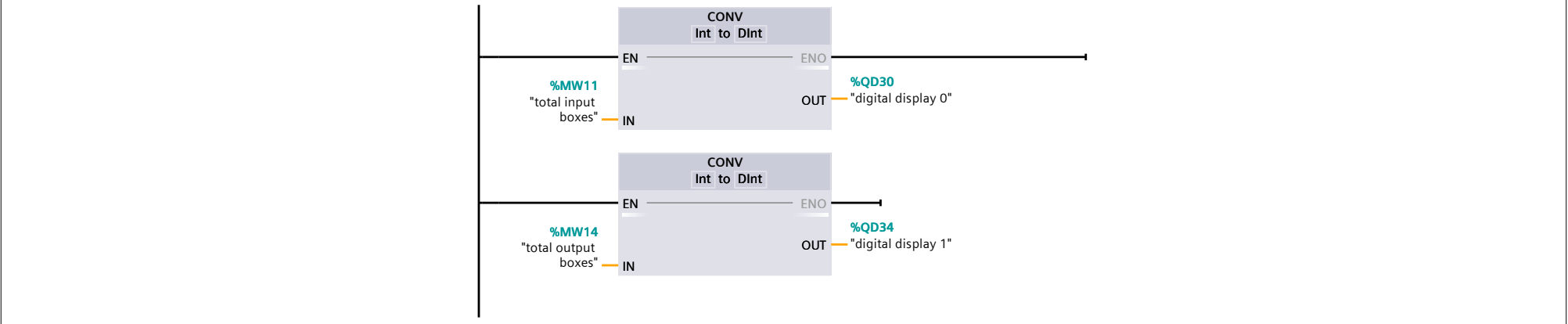
Network 7: counter input



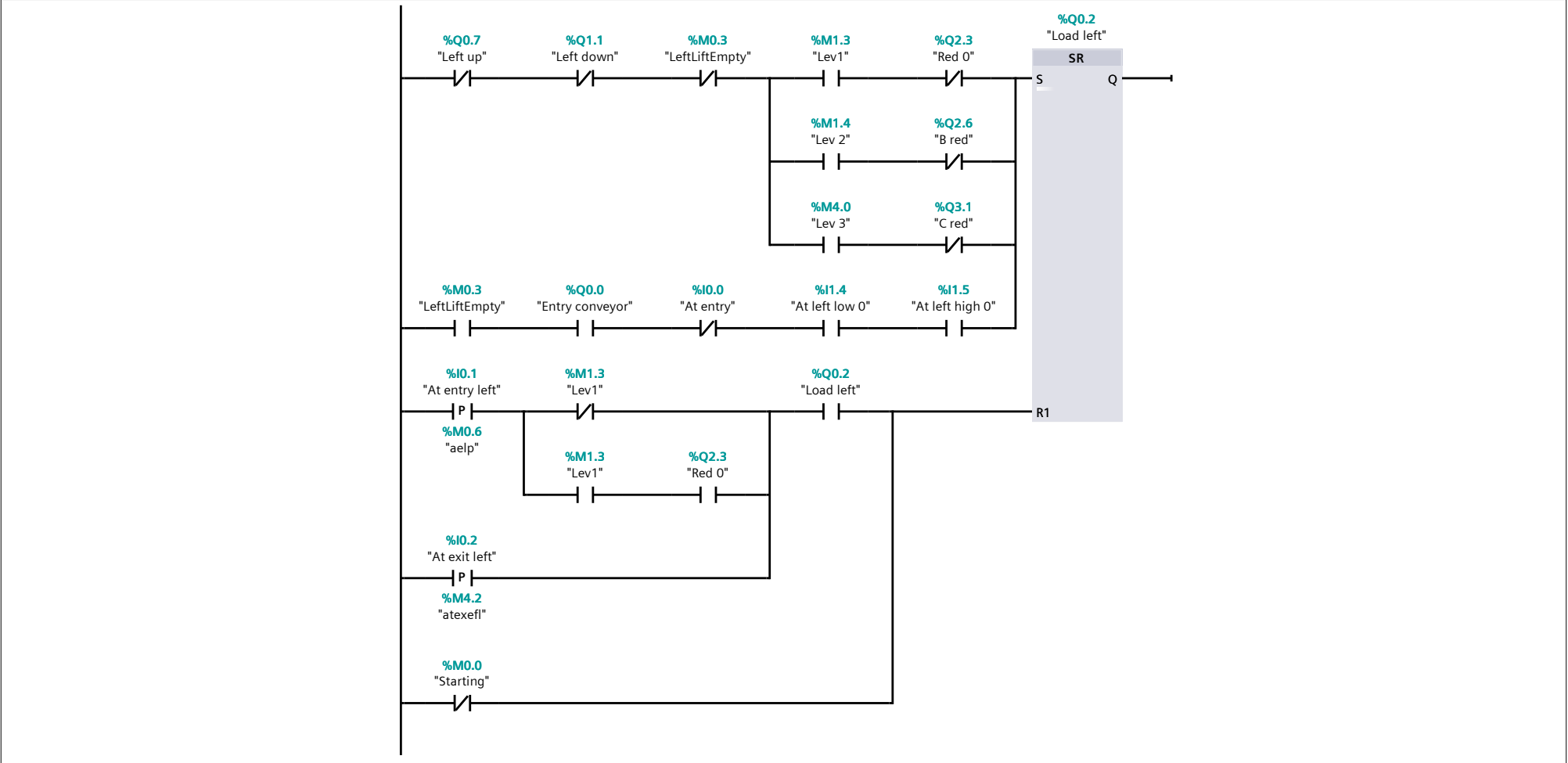
Network 8: counter output



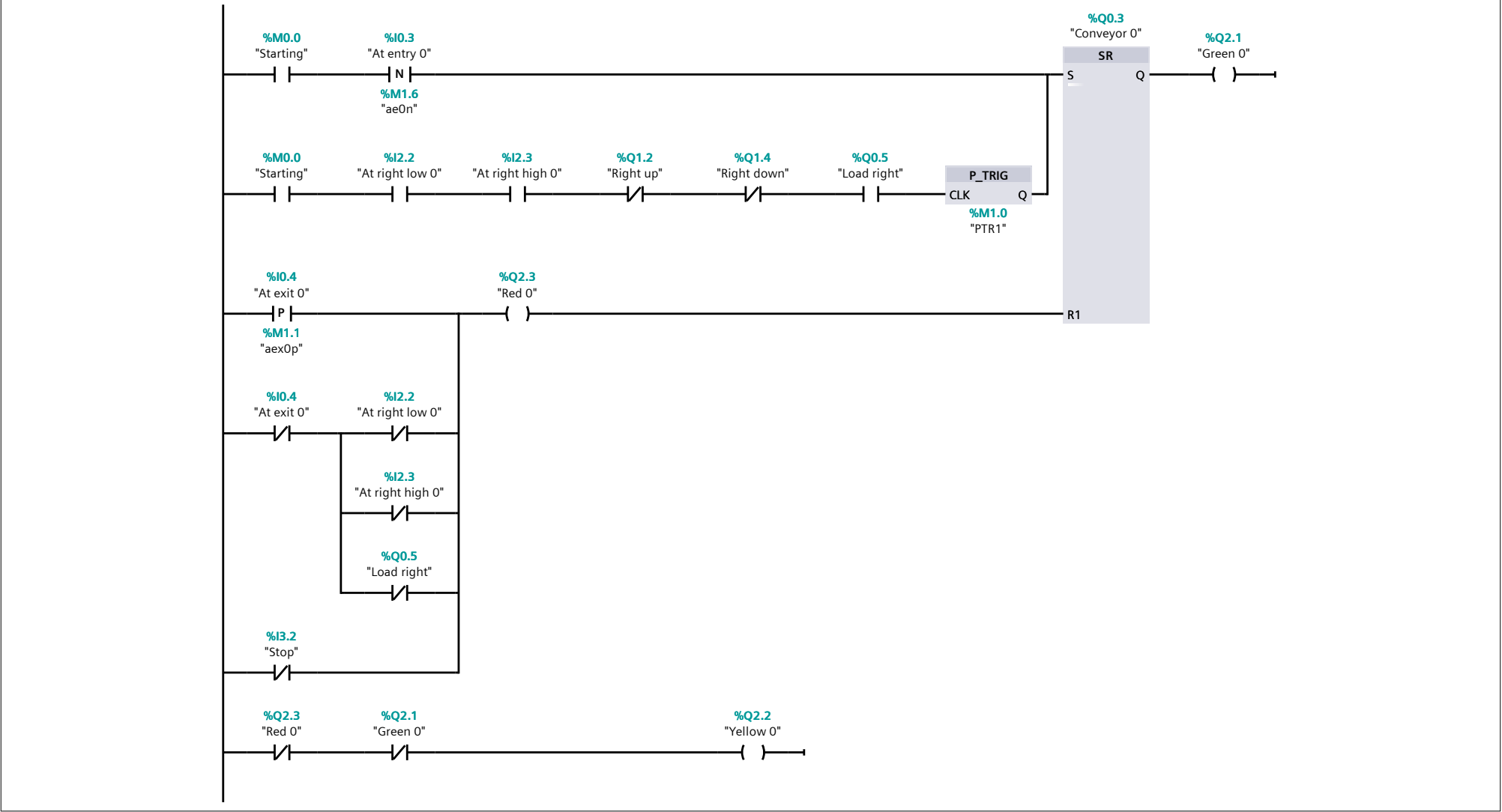
Network 9: Data type conversion



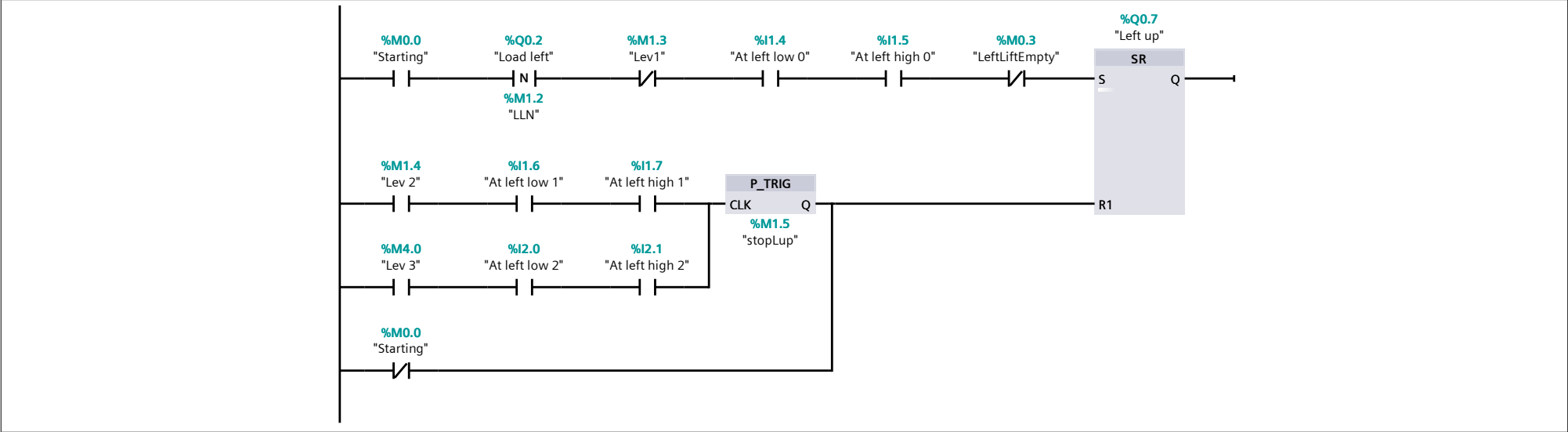
Network 10: Load left



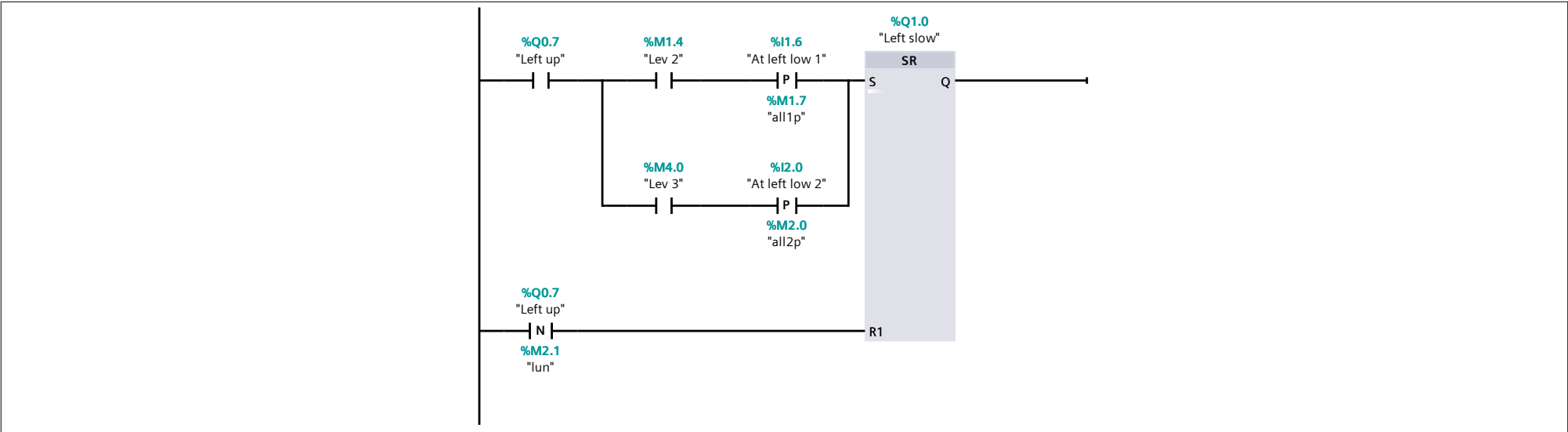
Network 11: Conveyor 0



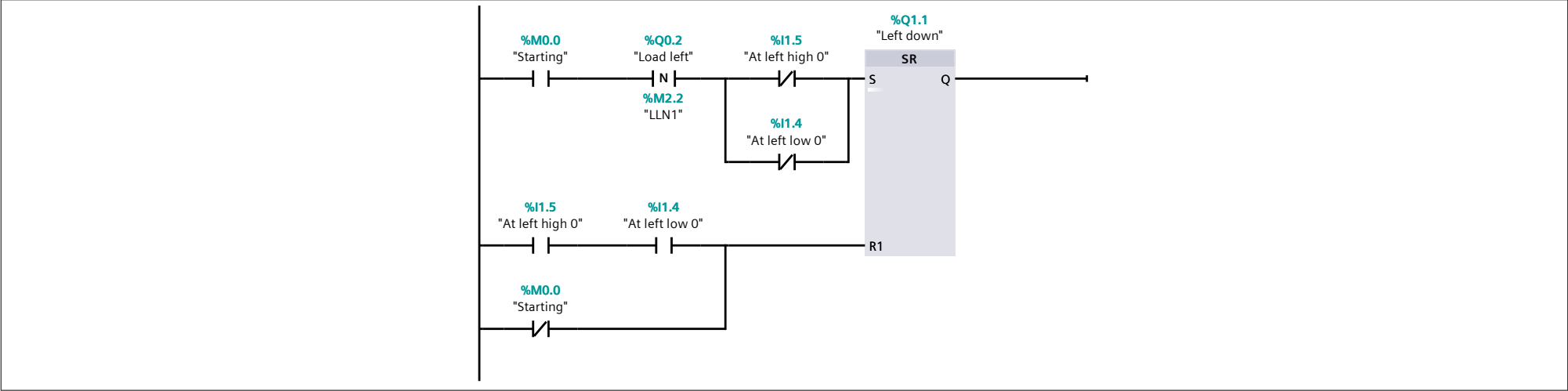
Network 12: Left up



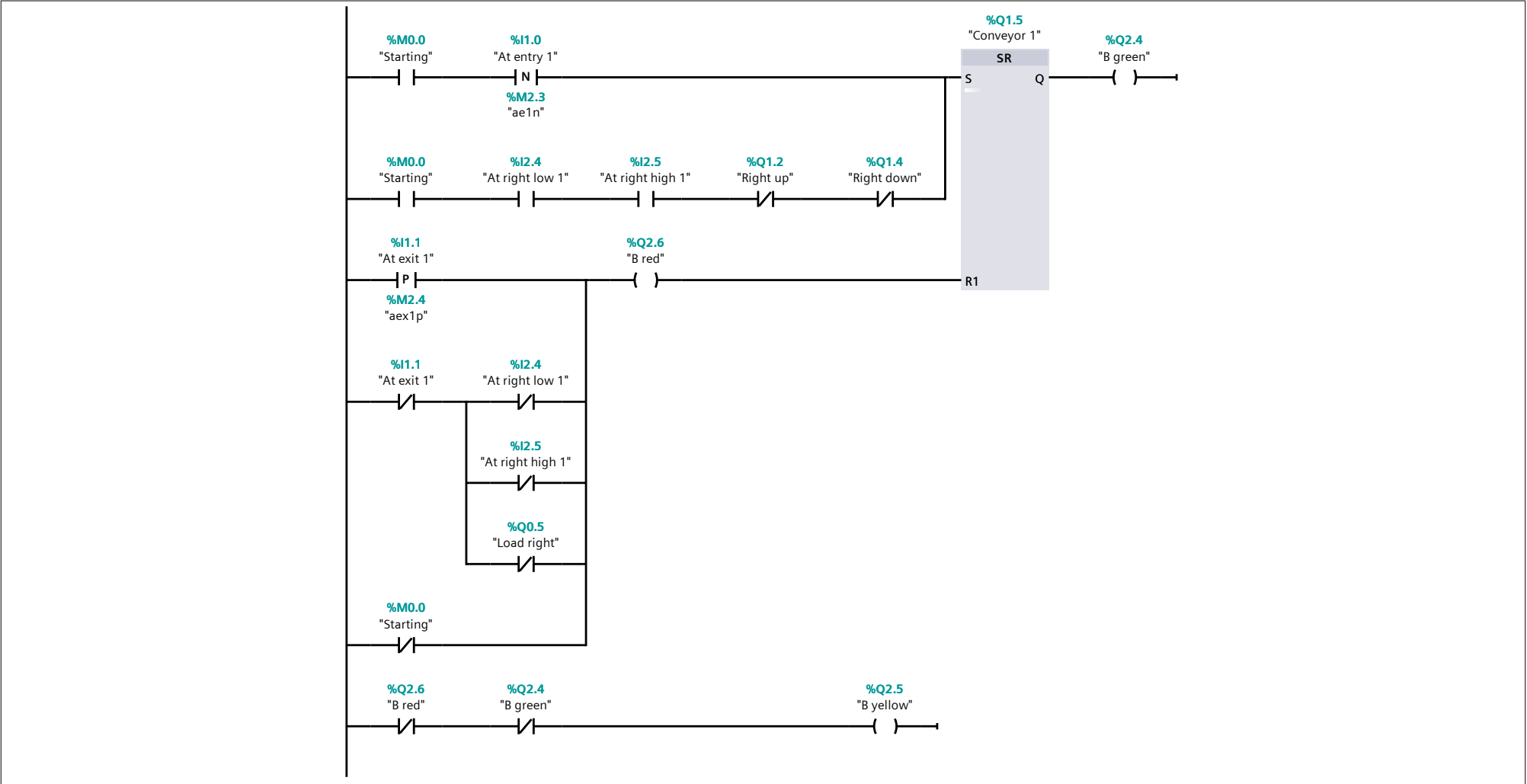
Network 13: Left slow



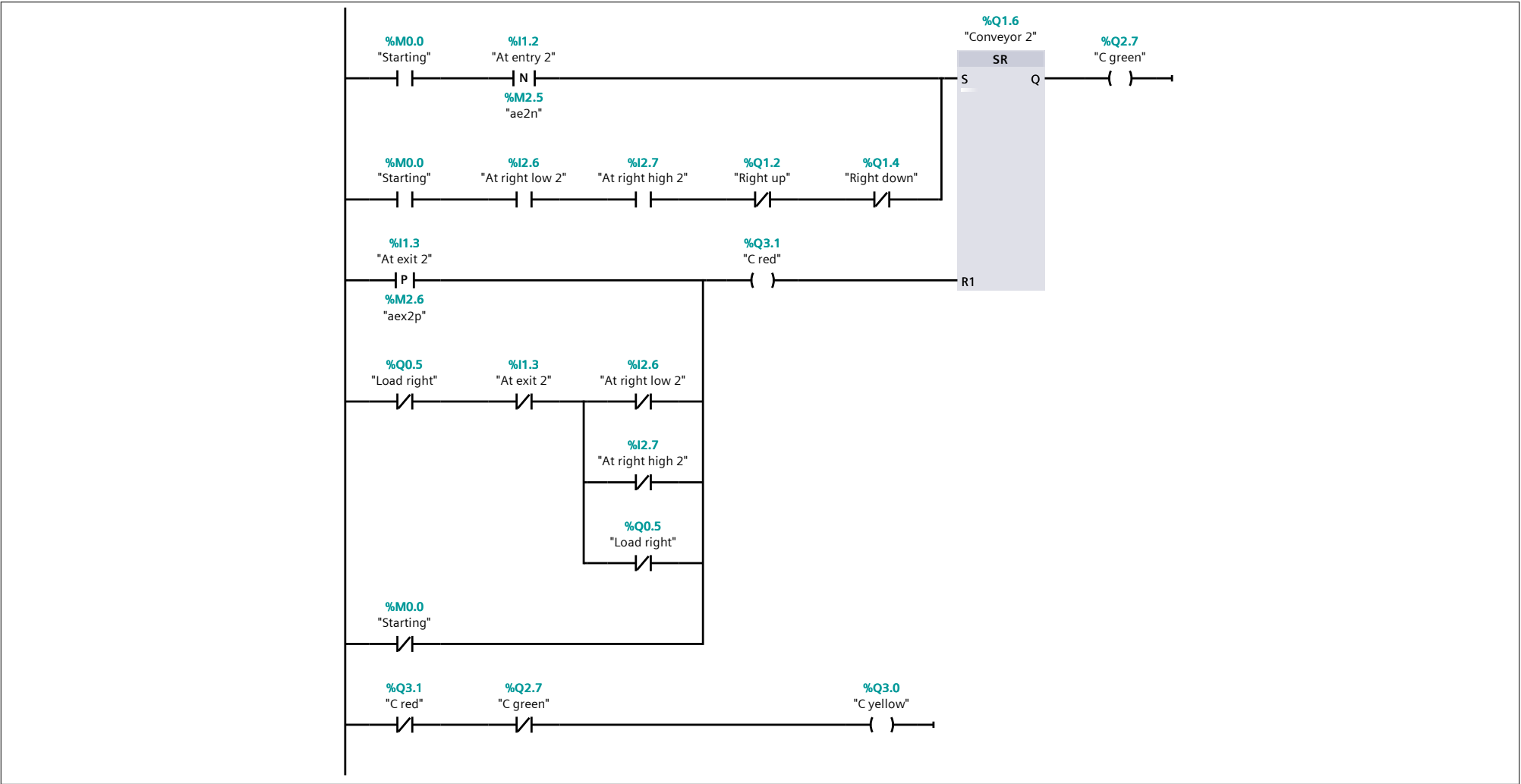
Network 14: Left Down



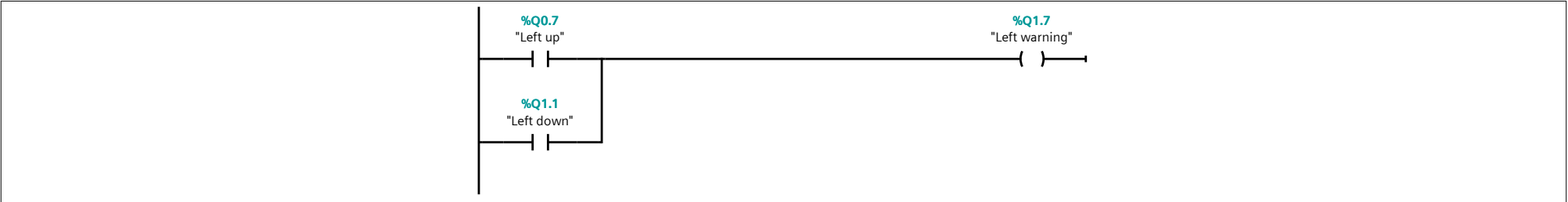
Network 15: Conveyor 1



Network 16: Conveyor 2

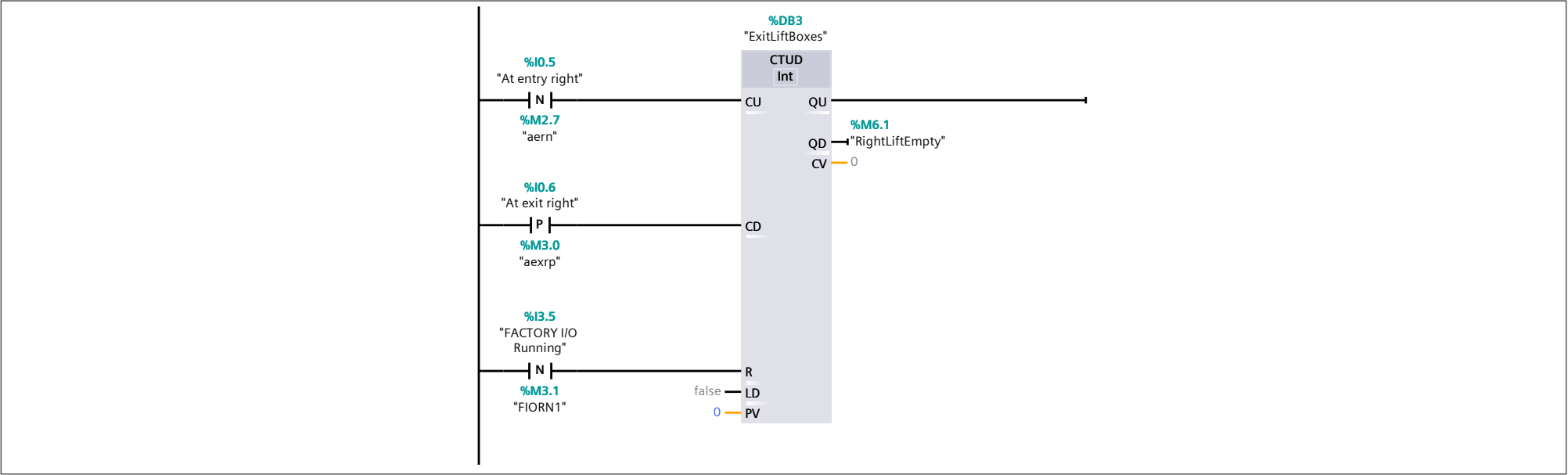


Network 17: Left Warning

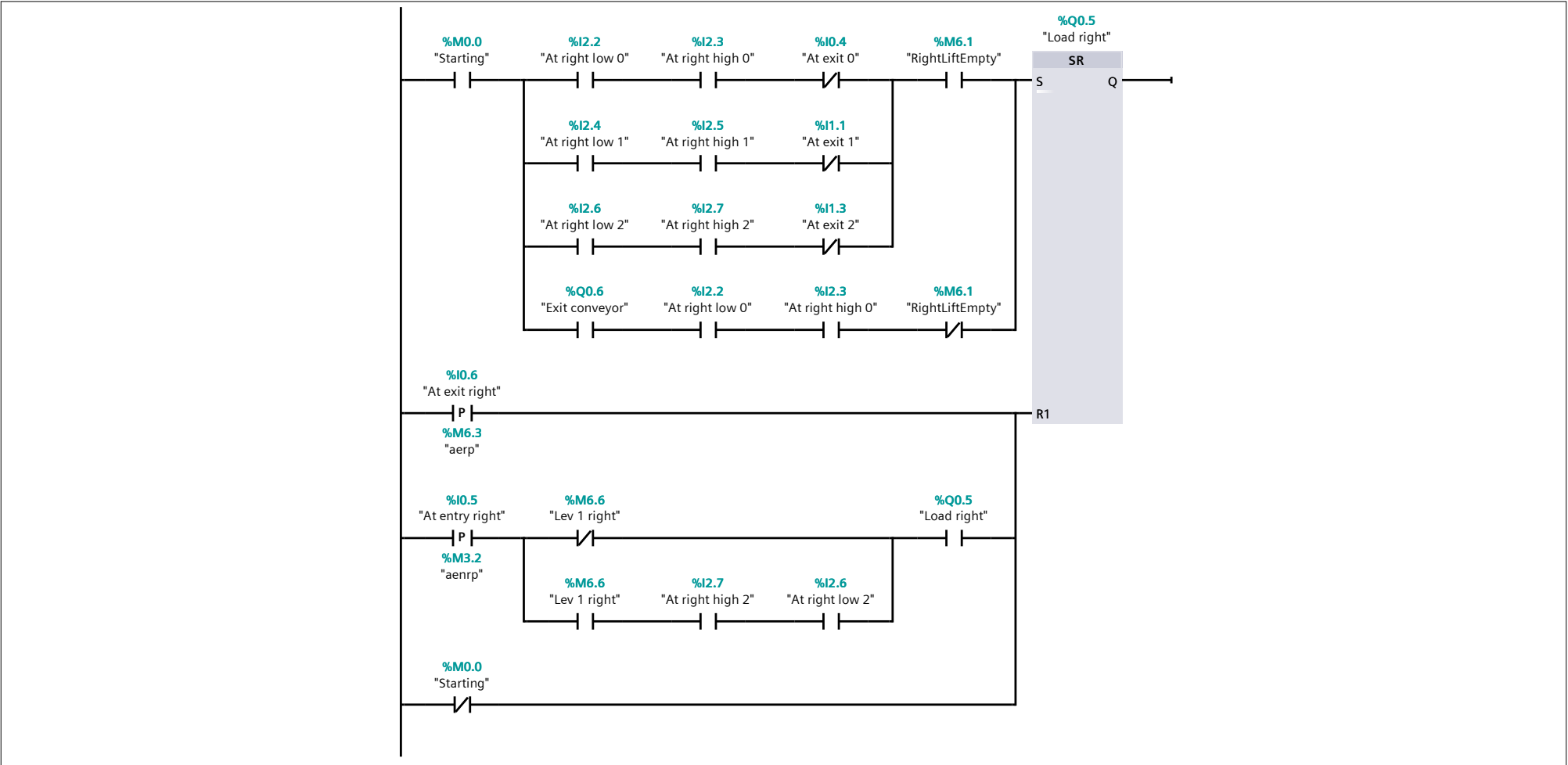


Network 18: Detecting box presence on exit lift

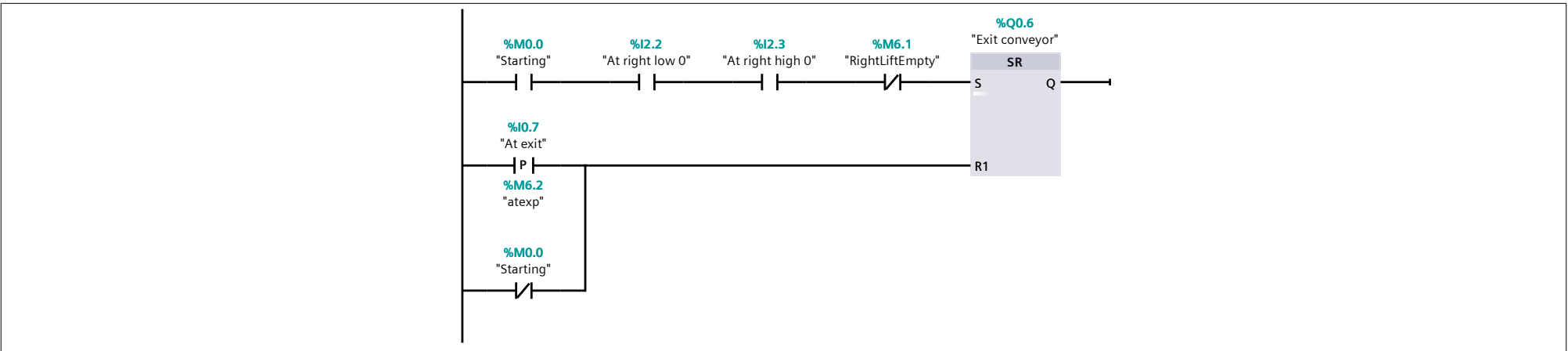




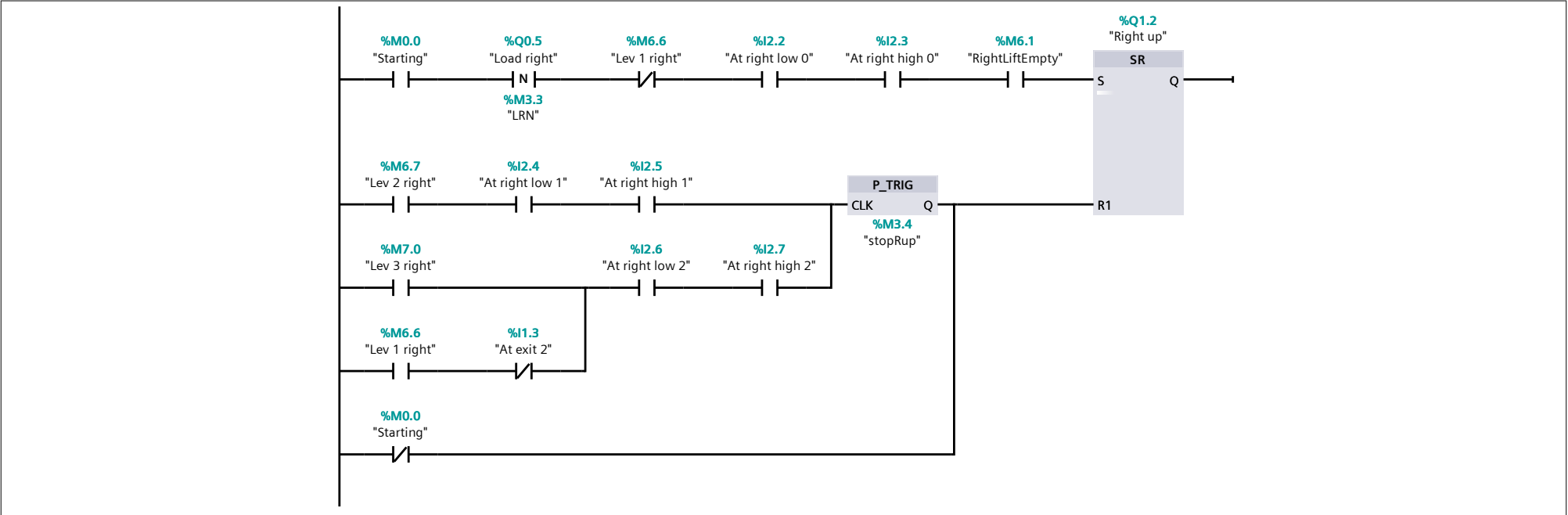
Network 19: Load right



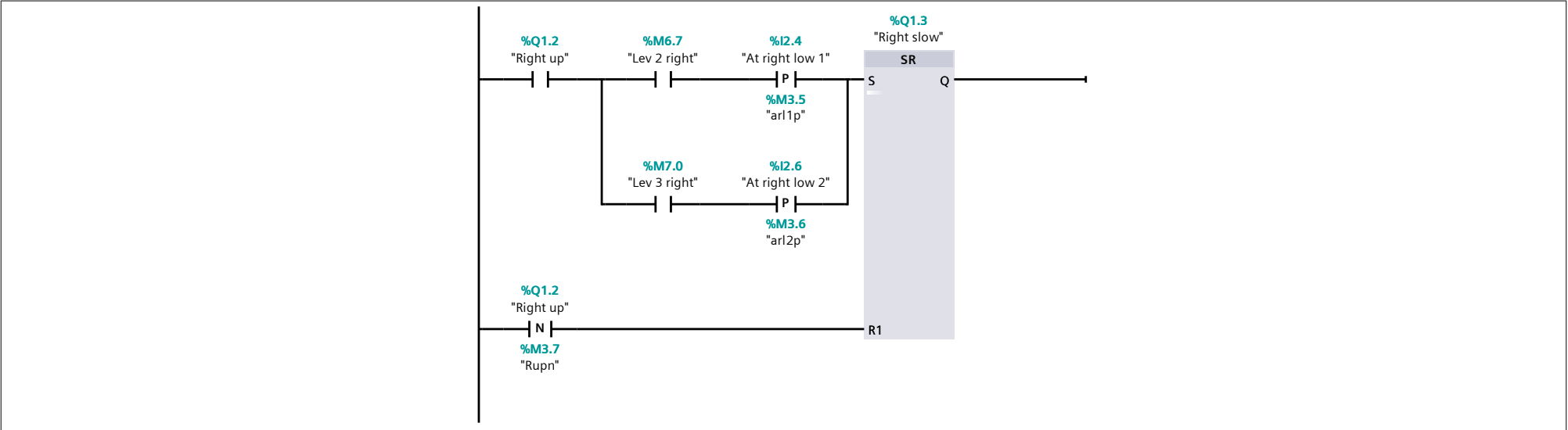
Network 20: Exit Conveyor



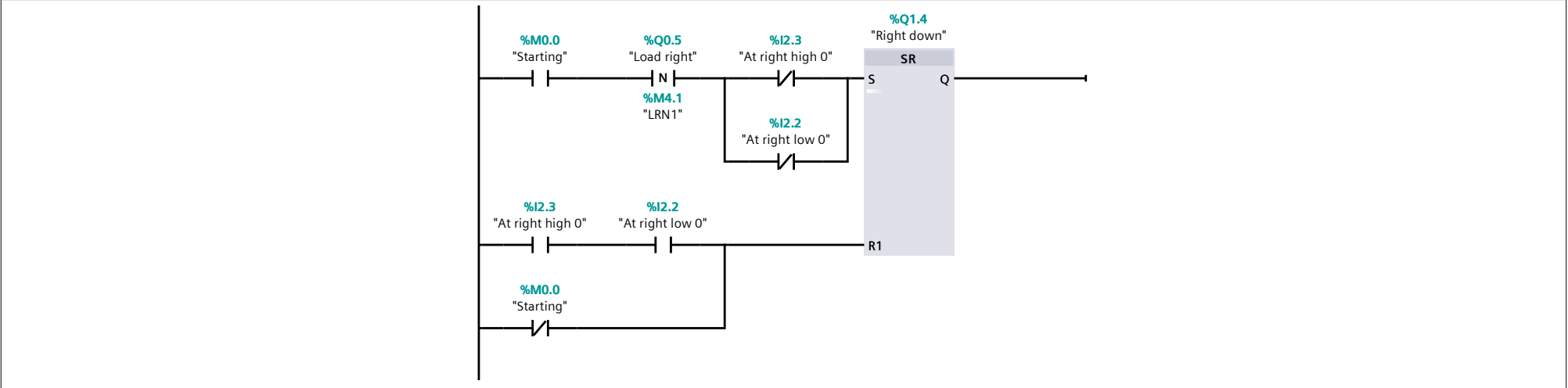
Network 21: Right up



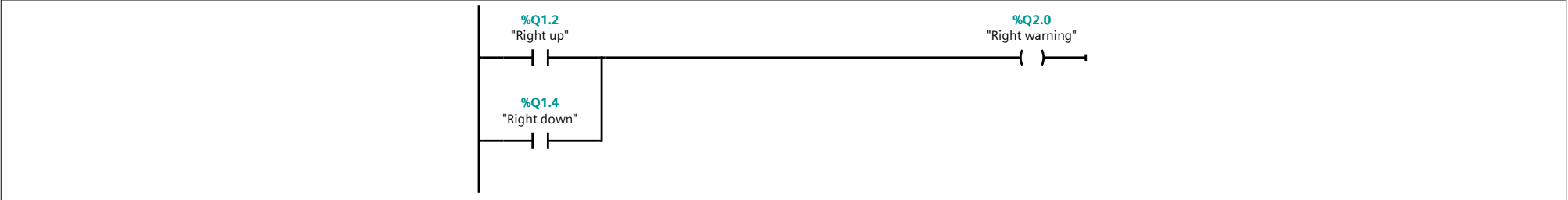
Network 22: Right slow



Network 23: Right Down



Network 24: Right Warning



Chapter 2: PLC Tags Mapping (I/O & Memory)

The following documentation table details the digital inputs, digital outputs, and internal memory addresses mapped within the PLC environment to coordinate the entire factory cell:

Tag Name	Logical Address	Data Type	Engineering Function / Description
Start	%I3.0	Bool	Master system activation push-button (Normally Open)
Stop	%I3.2	Bool	Standard system halt push-button (Normally Closed for fail-safe)
Emergency stop	%I3.3	Bool	Immediate power/control cutoff circuit (Normally Closed safety)
Starting	%M0.0	Bool	Master system latch bit indicating active operational state
Entry conveyor	%Q0.0	Bool	Command output to drive the primary entry feeding conveyor motor
At entry	%I10.0	Bool	Photoelectric sensor detecting box arrival at the entry check
LeftLiftEmpty	%M0.3	Bool	Internal flag verifying that the Left Lift is ready to receive
Left up	%Q0.7	Bool	High-speed upward movement command for Left Lift motor
Left slow	%Q1.0	Bool	De-acceleration/creeping speed command for Left Lift approach
Left down	%Q1.1	Bool	Downward return

Tag Name	Logical Address	Data Type	Engineering Function / Description
			movement command for Left Lift motor
Load left	%Q0.2	Bool	Internal deck conveyor of Left Lift for package intake/discharge
Right up	%Q1.2	Bool	High-speed upward movement command for Right Lift motor
Right down	%Q1.4	Bool	Downward return movement command for Right Lift motor
Right slow	%Q1.3	Bool	De-acceleration/creeping speed command for Right Lift approach
Load right	%Q0.5	Bool	Internal deck conveyor of Right Lift for package intake/discharge
Exit conveyor	%Q0.6	Bool	Command output to drive the final packaging clearance conveyor
total input boxes	%MW11	Int	Memory word storing total count of items entering the process
total output boxes	%MW14	Int	Memory word storing total count of items successfully cleared
digital display 0	%QD30	Dint	Double word output mapping values to the Input Count LED Screen
digital display 1	%QD34	Dint	Double word output

Tag Name	Logical Address	Data Type	Engineering Function / Description
			mapping values to the Output Count LED Screen

Chapter 3: Network-by-Network Functional Analysis

The Main Program Block (OB1) comprises 24 structured Networks executing sequentially. Below is the technical breakdown of each network:

Network 1: System Initialization

Establishes communication parameters and interfaces the memory space of the Siemens S7-1500 controller with the external Factory I/O driver module, ensuring synchronous execution loop updates without configuration faults.

Network 2 & 3: Sequence of Levels (Left & Right)

Invokes specific sequence Data Blocks (Sequence 1,2,3 Left DB & Right DB). These blocks act as absolute position trackers. They register real-time limits and proximity transitions to store the precise floor location of both lifts, preventing conflict commands.

Network 4: Master Start/Stop & Emergency Latch

Implements the master system latch using an SR bistable logic block. Momentarily pulsing the Start push-button (%I3.0) sets the "Starting" marker (%M0.0) to High. The Stop (%I3.2) and Emergency Stop (%I3.3) inputs are evaluated via open rungs linked directly to the Reset terminal, guaranteeing that an open circuit drops the system out of operation instantly.

Network 5: Entry Conveyor Actuation Control

Controls the main feeding conveyor (%Q0.0). It runs only when the system state is active ("Starting" = 1) and the Left Lift carriage signals a clear status via the "LeftLiftEmpty" flag (%M0.3). It cuts off immediately when an item hits the entry limit to avoid product jam conditions.

Network 6: Left Elevator Box Presence Tracking

Monitors internal platform sensors within the Left Lift. It generates the logic state for the "LeftLiftEmpty" memory marker (%M0.3) to authorize incoming transfers or initiate vertical travel commands.

Network 7 & 8: Input & Output Material Counters

Manages production statistics. Network 7 reads a dedicated Up-Counter (CTU) triggered by the input photo-eye "At entry" (%I10.0), updating total input records in %MW11. Network 8 tracks an independent CTU driven by the exit path photo-eye "At exit" (%I10.7), writing totals directly into %MW14.

Network 9: Data Type Conversion for Digital Displays

Resolves data structure type mismatches. Since CTU block accumulations produce standard 16-bit Integers (INT), and the multi-segment physical LED display panels require 32-bit Double Integers (DINT) mapped at %QD30 and %QD34, inline CONV blocks transform data syntax dynamically every program scan cycle.

Network 10 & 11: Loading Mechanism Left & Conveyor 0 Logic

Network 10 drives the onboard conveyor roller "Load left" (%Q0.2) to draw packages into the lifting car. Network 11 handles the ground lateral bypass conveyor (Conveyor 0). It integrates positive edge transition detectors (P_TRIG) to catch the leading edge of passing cargo, ensuring single-pulse executions.

Network 12, 13 & 14: Left Elevator Motion Control

Coordinates the three physical movement vectors of the Left Lift carriage:

- **Network 12 (Left up):** Fires the rapid ascent command output (%Q0.7) when destination conditions match higher floors.
- **Network 13 (Left slow):** Activates creeping/slow speed (%Q1.0) moments before hitting target target levels to achieve perfect leveling accuracy.
- **Network 14 (Left down):** Drives the rapid descent command (%Q1.1) to return the carrier to the home floor position upon load discharging.

Network 15 & 16: Level 1 and Level 2 Horizontal Distribution Conveyors

Coordinates mid-level sorting lines Conveyor 1 (%Q1.5) and Conveyor 2 (%Q1.6). These networks guide parcels along horizontal buffers once they exit the first lift, routing them towards the Right Lift reception points.

Network 17: Left Elevator Warning Signaling

An industrial safety network. It compiles active Left Up or Left Down states through a logical OR gate to activate the warning light output "Left warning" (%Q1.7), alerting floor operators of automated mechanical movement.

Network 18 & 19: Exit Lift Presence & Loading Logic

Network 18 checks parcel queuing on the secondary transfer junctions to toggle the "RightLiftEmpty" flag (%M6.1). Network 19 executes the onboard tracking deck "Load right" (%Q0.5) to lock arriving elements onto the right elevator structure safely.

Network 20: Final Stage Exit Conveyor Actuation

Drives the final discharge transport motor "Exit conveyor" (%Q0.6) as soon as the Right Lift lines up with the ground discharge station and finishes item transfer operations.

Network 21, 22 & 23: Right Elevator Motion Control

Directs the vertical drive components assigned to the Right sorting lift assembly:

- **Network 21 (Right up):** Drives the upward rapid travel command (%Q1.2) to collect items from upper accumulation tracks.
- **Network 22 (Right slow):** Deploys crawling speed approach (%Q1.3) to ensure precise mechanical alignment with upper structural shelves.
- **Network 23 (Right down):** Drives the high-speed downward command (%Q1.4) to rapidly lower loaded pallets toward the primary exit tracks.

Network 24: Right Elevator Warning Signaling

Operates the independent visual/audible safety alarm output "Right warn" (%Q2.0). It tracks the combined active travel expressions of the Right Lift motor using OR logic to prevent workspace hazards.

Chapter 4: Industrial Interlocking, Safety, and Fail-Safe Mechanisms

To ensure protection against mechanical stress, electrical overloads, or asynchronous signal evaluation errors, the program embeds specific safety layers:

1. **Software Motor Interlocking:** High-voltage bi-directional induction motors used in automated lifters must never be energized in opposite directions simultaneously, as this

causes phase-to-phase short circuits. The ladder logic integrates a interlocking scheme where a normally closed contact of the Down output is wired in series with the Up output line, and vice versa. This software block makes simultaneous activation physically impossible inside the CPU image memory.

2. **Fail-Safe Stop Configuration:** Hardware control devices like Stop buttons and E-Stops maintain continuous electrical continuity when untouched. The PLC interprets this steady input state as a logical high (1). By programming these markers as normally open contacts inside the logic rungs leading to critical resets, any real-world wire break, sensor failure, or active operator press drops the line state instantly to low (0), executing a comprehensive system trip.
3. **Two-Speed Micro-Positioning Interlock:** Rather than forcing structural mechanisms to stop instantly from a high linear velocity, the control loop engages slow-speed transitions. This cushions deceleration forces, minimizing structural vibration, preventing cargo slippage, and protecting the mechanical integrity of chains and drive gears.

[illegible]

28	Lev 3	Standard-\ Bool	%M4.0		True	True	True						
29	Red 0	Standard-\ Bool	%Q2.3		True	True	True						
30	B red	Standard-\ Bool	%Q2.6		True	True	True						
31	C red	Standard-\ Bool	%Q3.1		True	True	True						
32	Load left	Standard-\ Bool	%Q0.2		True	True	True						
33	At entry 0	Standard-\ Bool	%I0.3		True	True	True						
34	ae0n	Standard-\ Bool	%M1.6		True	True	True						
35	Conveyor	Standard-\ Bool	%Q0.3		True	True	True						
36	Green 0	Standard-\ Bool	%Q2.1		True	True	True						
37	At right lov	Standard-\ Bool	%I2.2		True	True	True						
38	At right hig	Standard-\ Bool	%I2.3		True	True	True						
39	Right up	Standard-\ Bool	%Q1.2		True	True	True						
40	Right down	Standard-\ Bool	%Q1.4		True	True	True						
41	Load right	Standard-\ Bool	%Q0.5		True	True	True						
42	PTR1	Standard-\ Bool	%M1.0		True	True	True						
43	At exit 0	Standard-\ Bool	%I0.4		True	True	True						
44	aex0p	Standard-\ Bool	%M1.1		True	True	True						
45	Yellow 0	Standard-\ Bool	%Q2.2		True	True	True						
46	LLN	Standard-\ Bool	%M1.2		True	True	True						
47	At left low	Standard-\ Bool	%I1.6		True	True	True						
48	At left high	Standard-\ Bool	%I1.7		True	True	True						
49	stopLup	Standard-\ Bool	%M1.5		True	True	True						
50	At left low	Standard-\ Bool	%I2.0		True	True	True						
51	At left high	Standard-\ Bool	%I2.1		True	True	True						
52	all1p	Standard-\ Bool	%M1.7		True	True	True						
53	all2p	Standard-\ Bool	%M2.0		True	True	True						
54	Left slow	Standard-\ Bool	%Q1.0		True	True	True						

55	Jun	Standard-\ Bool	%M2.1		True	True	True						
56	LLN1	Standard-\ Bool	%M2.2		True	True	True						
57	At entry 1	Standard-\ Bool	%I1.0		True	True	True						
58	ae1n	Standard-\ Bool	%M2.3		True	True	True						
59	Conveyor	Standard-\ Bool	%Q1.5		True	True	True						
60	B green	Standard-\ Bool	%Q2.4		True	True	True						
61	At right lov	Standard-\ Bool	%I2.4		True	True	True						
62	At right hig	Standard-\ Bool	%I2.5		True	True	True						
63	At exit 1	Standard-\ Bool	%I1.1		True	True	True						
64	aex1p	Standard-\ Bool	%M2.4		True	True	True						
65	B yellow	Standard-\ Bool	%Q2.5		True	True	True						
66	At entry 2	Standard-\ Bool	%I1.2		True	True	True						
67	ae2n	Standard-\ Bool	%M2.5		True	True	True						
68	At right lov	Standard-\ Bool	%I2.6		True	True	True						
69	At right hig	Standard-\ Bool	%I2.7		True	True	True						
70	Conveyor	Standard-\ Bool	%Q1.6		True	True	True						
71	C green	Standard-\ Bool	%Q2.7		True	True	True						
72	At exit 2	Standard-\ Bool	%I1.3		True	True	True						
73	aex2p	Standard-\ Bool	%M2.6		True	True	True						
74	C yellow	Standard-\ Bool	%Q3.0		True	True	True						
75	Left warni	Standard-\ Bool	%Q1.7		True	True	True						
76	At entry rig	Standard-\ Bool	%I0.5		True	True	True						
77	aern	Standard-\ Bool	%M2.7		True	True	True						
78	At exit righ	Standard-\ Bool	%I0.6		True	True	True						
79	aexrp	Standard-\ Bool	%M3.0		True	True	True						
80	FIORN1	Standard-\ Bool	%M3.1		True	True	True						
81	RightLiftEr	Standard-\ Bool	%M6.1		True	True	True						

	1	2	3	4	5	6	7	8	9	10	11	12
81	RightLiftEr	Standard-\ Bool	%M6.1			True	True	True				
82	Exit conve	Standard-\ Bool	%Q0.6			True	True	True				
83	aerp	Standard-\ Bool	%M6.3			True	True	True				
84	aenrp	Standard-\ Bool	%M3.2			True	True	True				
85	Lev 1 right	Standard-\ Bool	%M6.6			True	True	True				
86	atexp	Standard-\ Bool	%M6.2			True	True	True				
87	At exit	Standard-\ Bool	%I0.7			True	True	True				
88	LRN	Standard-\ Bool	%M3.3			True	True	True				
89	Lev 2 right	Standard-\ Bool	%M6.7			True	True	True				
90	Lev 3 right	Standard-\ Bool	%M7.0			True	True	True				
91	stopRup	Standard-\ Bool	%M3.4			True	True	True				
92	arl1p	Standard-\ Bool	%M3.5			True	True	True				
93	arl2p	Standard-\ Bool	%M3.6			True	True	True				
94	Right slow	Standard-\ Bool	%Q1.3			True	True	True				
95	Rupn	Standard-\ Bool	%M3.7			True	True	True				
96	LRN1	Standard-\ Bool	%M4.1			True	True	True				
97	Right warn	Standard-\ Bool	%Q2.0			True	True	True				
98	Reset	Standard-\ Bool	%I3.1			True	True	True				
99	Unload lef	Standard-\ Bool	%Q0.1			True	True	True				
100	Unload rig	Standard-\ Bool	%Q0.4			True	True	True				
101	Reset light	Standard-\ Bool	%Q3.3			True	True	True				
102	atexefl	Standard-\ Bool	%M4.2			True	True	True				
103	PC0	Standard-\ Bool	%M4.3			True	True	True				
104	PC1	Standard-\ Bool	%M4.4			True	True	True				
105	PC2	Standard-\ Bool	%M4.5			True	True	True				
106	digital disp	Standard-\ DInt	%QD30			True	True	True				
107	total input	Standard-\ Int	%MW11			True	True	True				

104	PC1	Standard-\ Bool	%M4.4			True	True	True
105	PC2	Standard-\ Bool	%M4.5			True	True	True
106	digital disp	Standard-\ DInt	%QD30			True	True	True
107	total input	Standard-\ Int	%MW11			True	True	True
108	total output	Standard-\ Int	%MW14			True	True	True
109	boxes insic	Standard-\ Int	%MW16			True	True	True
110	digital disp	Standard-\ DInt	%QD34			True	True	True



SENSORS

At entry
At entry 0
At entry 1
At entry 2
At entry left
At entry right
At exit
At exit 0
At exit 1
At exit 2
At exit left
At exit right
At left high 0
At left high 1
At left high 2
At left low 0
At left low 1
At left low 2
At right high 0
At right high 1
At right high 2
At right low 0
At right low 1
At right low 2
Auto
Emergency stop
FACTORY I/O (Paused)
FACTORY I/O (Reset)
FACTORY I/O (Running)
FACTORY I/O (Time Scale)
Manual
Reset
Start

At entry	%I0.0	%Q0.0	Entry conveyor
At entry left	%I0.1	%Q0.1	Unload left
At exit left	%I0.2	%Q0.2	Load left
At entry 0	%I0.3	%Q0.3	Conveyor 0
At exit 0	%I0.4	%Q0.4	Unload right
At entry right	%I0.5	%Q0.5	Load right
At exit right	%I0.6	%Q0.6	Exit conveyor
At exit	%I0.7	%Q0.7	Left up
At entry 1	%I1.0	%Q1.0	Left slow
At exit 1	%I1.1	%Q1.1	Left down
At entry 2	%I1.2	%Q1.2	Right up
At exit 2	%I1.3	%Q1.3	Right slow
At left low 0	%I1.4	%Q1.4	Right down
At left high 0	%I1.5	%Q1.5	Conveyor 1
At left low 1	%I1.6	%Q1.6	Conveyor 2
At left high 1	%I1.7	%Q1.7	Left warning
At left low 2	%I2.0	%Q2.0	Right warning
At left high 2	%I2.1	%Q2.1	Green 0
At right low 0	%I2.2	%Q2.2	Yellow 0
At right high 0	%I2.3	%Q2.3	Red 0
At right low 1	%I2.4	%Q2.4	B green
At right high 1	%I2.5	%Q2.5	B yellow
At right low 2	%I2.6	%Q2.6	B red
At right high 2	%I2.7	%Q2.7	C green
Start	%I3.0	%Q3.0	C yellow
Reset	%I3.1	%Q3.1	C red
Stop	%I3.2	%Q3.2	Start light
Emergency stop	%I3.3	%Q3.3	Reset light
Auto	%I3.4	%Q3.4	Stop light
FACTORY I/O (Running)	%I3.5	%Q3.5	
	%I3.6	%Q3.6	
	%I3.7	%Q3.7	
	%I4.0	%Q4.0	
	%I4.1	%Q4.1	

ACTUATORS

B green
B red
B yellow
C green
C red
C yellow
Conveyor 0
Conveyor 1
Conveyor 2
Digital Display 0
Digital Display 1
Emitter
Entry conveyor
Exit conveyor
FACTORY I/O (Camera Position)
FACTORY I/O (Pause)
FACTORY I/O (Reset)
FACTORY I/O (Run)
Green 0
Left down
Left slow
Left up
Left warning
Load left
Load right
Red 0
Remover
Reset light
Right down
Right slow
Right up
Right warning
Start light

FACTORY I/O (Time Scale)

%I7.6

%Q7.6

%I7.7

%Q7.7

%ID30 (REAL)

(DINT) %QD30

Digital Display 0

%ID34

(DINT) %QD34

Digital Display 1

%ID38

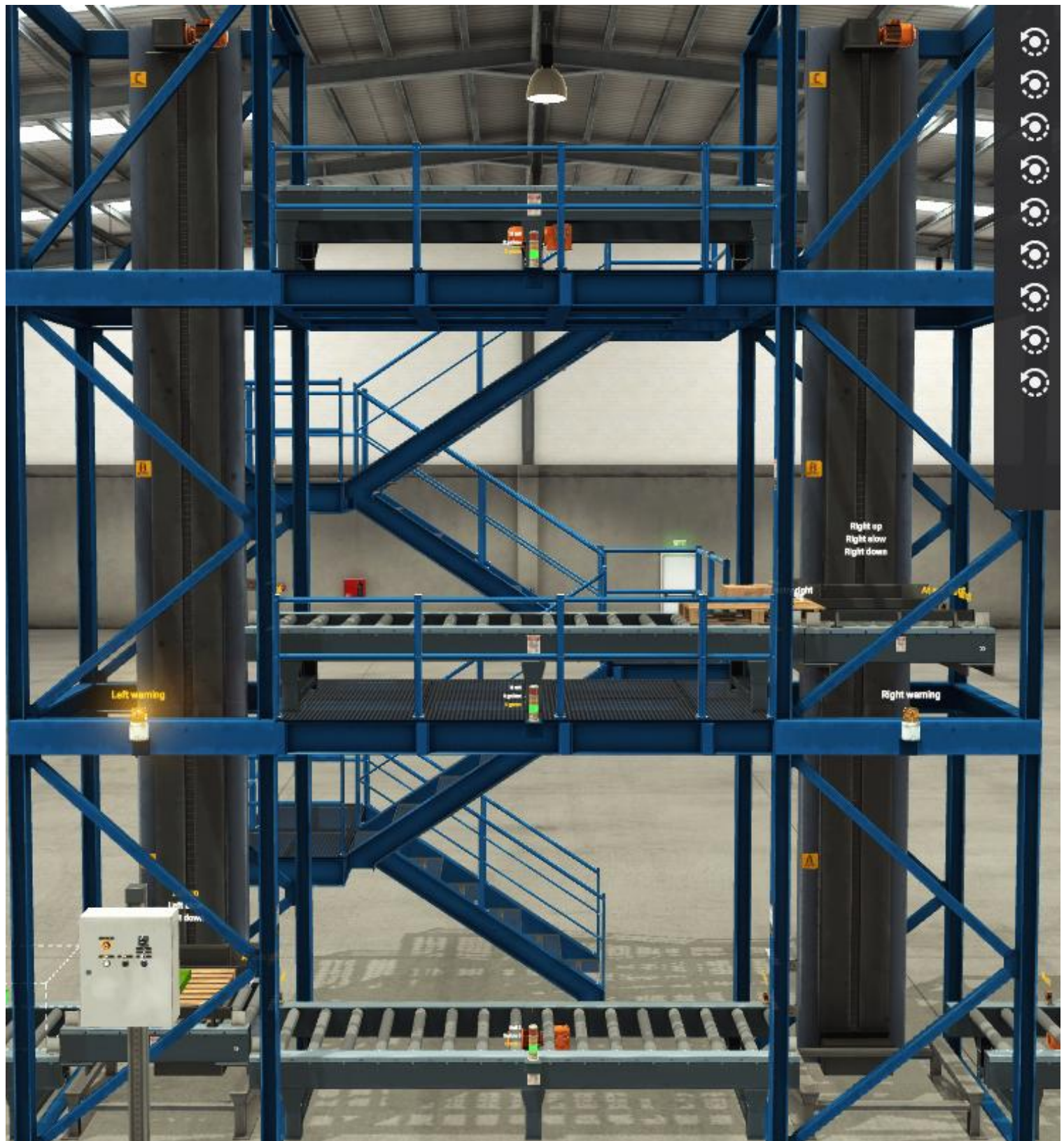
%QD38

%ID42

%QD42









Siemens

C_1 [CPU 1511-1 PN]

SIEMENS

RUN / STOP

ERROR

MAINT

RUN

STOP

PAUSE

MRES

X1192.168.0.1

no project>



